Reg. No.															

M.Tech. (Working Professionals) DEGREE EXAMINATION, JANUARY 2025

First Semester

21CSC555J - MACHINE LEARNING ALGORITHMS

(For the candidates admitted from the academic year2024 - 2025)

Time: 3 hours | Max. Marks: 100

(5 x 20 = 100 Marks)
Answer **ANY FIVE** Questions

	Marks	BL	CO	PO
1. (i) Consider a dataset where the independent variable X represents the number of study hours, and the dependent variable Y represents exam scores: Explain the process of deriving the simple linear regression equation. Describe the mathematical steps to calculate the slope and intercept of the regression line. Interpret the regression line in the context of predicting exam scores based on study hours. (10 Marks) (ii) Define the concepts of overfitting and underfitting in the context of predicting students' academic performance. Discuss how they can impact the reliability of the prediction model. (5 Marks) (iii) Identify and explain commonly used evaluation metrics for regression models in the context of assessing prediction accuracy for student exam scores. Discuss why each metric is important for evaluating model performance. (5 Marks)	20	3	1	1
2. A retail analytics company is developing a machine learning model to predict customer purchasing behavior. The goal is to classify customers into three spending categories: "low spender," "moderate spender," and "high spender." The dataset includes features such as monthly income, number of purchases, and average purchase value. Explain how Logistic Regression could be applied to estimate the probability of a customer being a "high spender." Describe the process of converting the linear regression equation into a sigmoid function and interpreting the resulting probability values for classifying customers. Since there are three categories ("low spender," "moderate spender," and "high spender"), explain how Logistic Regression can be adapted for multi-class classification. Describe methods like One-vs-Rest (OvR) or Softmax Regression for handling multiple classes.	20	3	2	2
3. A telecommunications company is building a machine learning model to predict customer churn (whether a customer will switch to another service provider). The dataset includes customer demographics, usage patterns, and support interaction history. The team wants to ensure the model generalizes well to new data and avoids overfitting. Explain how cross-validation can be utilized to assess the performance of the churn prediction model and discuss the purpose of cross-validation and how it helps in improving the generalization of the model to unseen data.	20	4	2	1
4. A food delivery app is building a recommendation system to suggest restaurants to users based on preferences and location. The dataset includes features like cuisine type, user ratings, average cost, and proximity. The team decides to use the K-Nearest Neighbors (KNN) algorithm. Explain how the KNN algorithm works in this context and discuss how the choice of **K** affects model performance, including challenges with very small or large K values. Describe the role of distance metrics, comparing Euclidean and Manhattan distances, and recommend the most suitable metric for this scenario. Finally, suggest evaluation metrics to assess the model's effectiveness in recommending relevant restaurants, justifying their importance in this context.	20	3	3	2

5. Consider the car theft prediction problem with the following attributes: Color (Categorical: Red, Blue, Black, etc.), Type (Categorical: SUV, Sedan, Truck, etc.), Origin (Categorical: Domestic, Imported), and the target variable Stolen (Binary: Yes, No). Assume that no pair of features are dependent, and each feature has equal influence on the classification. For this task, describe the necessary preprocessing steps you would take before applying a classification algorithm to this dataset, including how you would handle categorical data and which encoding methods you would use. Then, suggest a suitable classification algorithm, justifying your choice based on the nature of the features and the target variable. After selecting an algorithm, implement the classification model and evaluate its performance using appropriate metrics. Finally, after training the model, predict whether a car with the following attributes (Color = Red, Type = SUV, Origin = Domestic) is stolen or not, and explain how the prediction is made and interpret the result. 20 4 3 2

Example No.	Color	Type	Origin	Stolen?
1	Red	Sports	Domestic	Yes
2	Red	Sports	Domestic	No
3	Red	Sports	Domestic	Yes
4	Yellow	Sports	Domestic	No
5	Yellow	Sports	Imported	Yes
6	Yellow	SUV	Imported	No
7	Yellow	SUV	Imported	Yes
8	Yellow	SUV	Domestic	No
9	Red	SUV	Imported	No
10	Red	Sports	Imported	Yes

6. A sports club wants to predict whether they should hold outdoor events based on weather conditions to improve turnout and satisfaction. They have historical data on past events, including details like outlook, temperature, humidity, wind, and whether the event was successful ("Yes" or "No"). Using this data, construct a decision tree model to help the sports club decide whether or not to hold an outdoor event. Describe the steps for selecting attributes, determining splitting criteria, and building the tree to reach a decision on whether they should "play" or "not play" under specific weather conditions. 20 3 4 1

Day	outlook	temperature	humidity	wind	Decision
1	sunny	hot	high	weak	No
2	sunny	hot	high	strong	No
3	overcast	hot	high	weak	Yes
4	rainfall	mild	high	weak	Yes
5	rainfall	cool	normal	weak	Yes
6	rainfall	cool	normal	strong	No
7	overcast	cool	normal	wtrong	Yes
8	sunny	mild	high	weak	No
9	sunny	cool	normal	weak	Yes
10	rainfall	mild	normal	weak	Yes
11	sunny	mild	normal	strong	Yes
12	overcast	mild	high	strong	Yes
13	overcast	hot	normal	weak	Yes
14	rainfall	mild	high	strong	No

7. Consider a dataset with two features: height (in cm) and weight (in kg) of individuals. You decide to use the K-means clustering algorithm to segment the dataset into two groups based on these features. What steps would you take before applying the algorithm, and how would you determine the optimal number of clusters (k) for this dataset? Discuss at least two methods to choose the value of k. After applying K-means, you find that Cluster 1 has a mean height of 160 cm and a mean weight of 55 kg, while Cluster 2 has a mean height of 180 cm and a mean weight of 75 kg. What can you infer about the individuals in each cluster? 20 4 5 2

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